Unknown property 'justify-content', Line 86, Column 13
Sublime Text 3
Ps
Pr
Ae

ER and EER

01 ER Diagrams

02 EER Diagrams

03 SQL

04 Assignment

Consider a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

1. List the strong (nonweak) entity types in the ER diagram.
2. Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
3. What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?
4. List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type.
5. List concisely the user requirements that led to this ER schema design.
6. Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1,000 loans. How does this show up on the (min, max) constraints?

Solution:

1. Entity types: BANK, ACCOUNT, CUSTOMER, LOAN
2. Weak entity type: BANK-BRANCH. Partial key: BranchNo. Identifying relationship: BRANCHES.
3. The partial key BranchNo in BANK-BRANCH specifies that the same BranchNo value may occur under different BANKs. The identifying relationship BRANCHES specifies that BranchNo values are uniquely assigned for those BANK-BRANCH entities that are related to the same BANK entity. Hence, the combination of BANK Code and BranchNo together constitute a full identifier for a BANK-BRANCH.

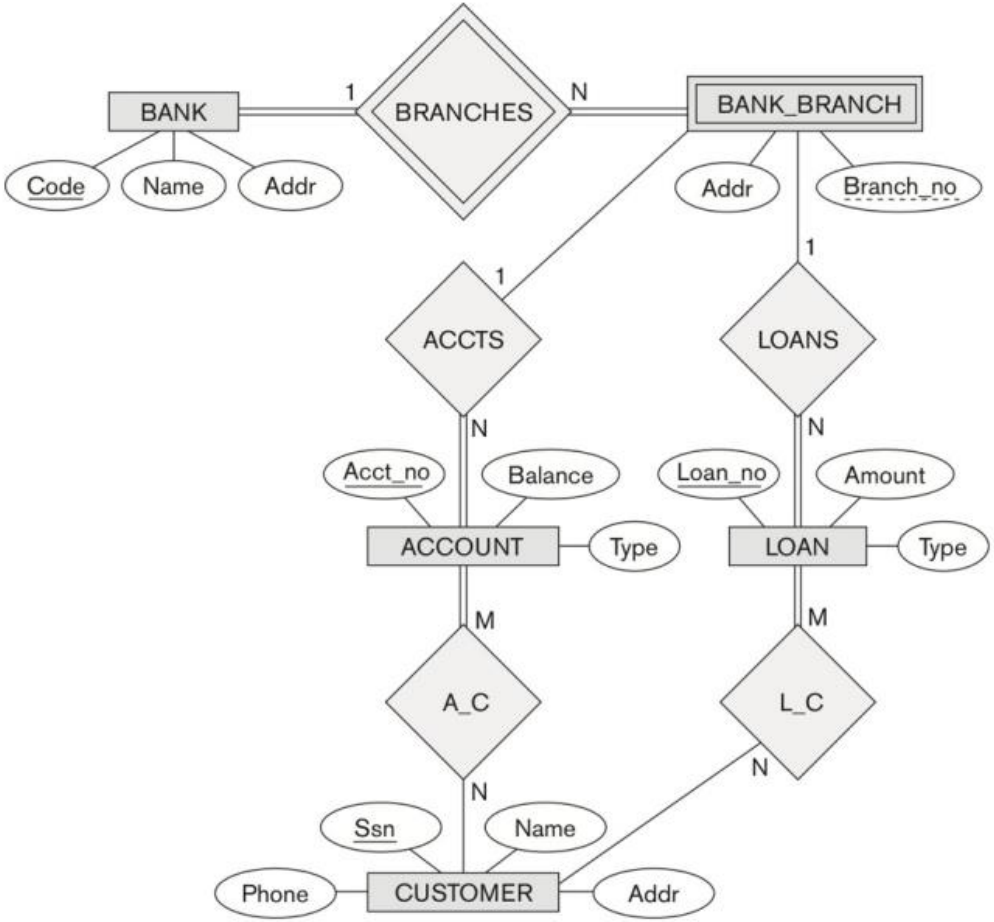
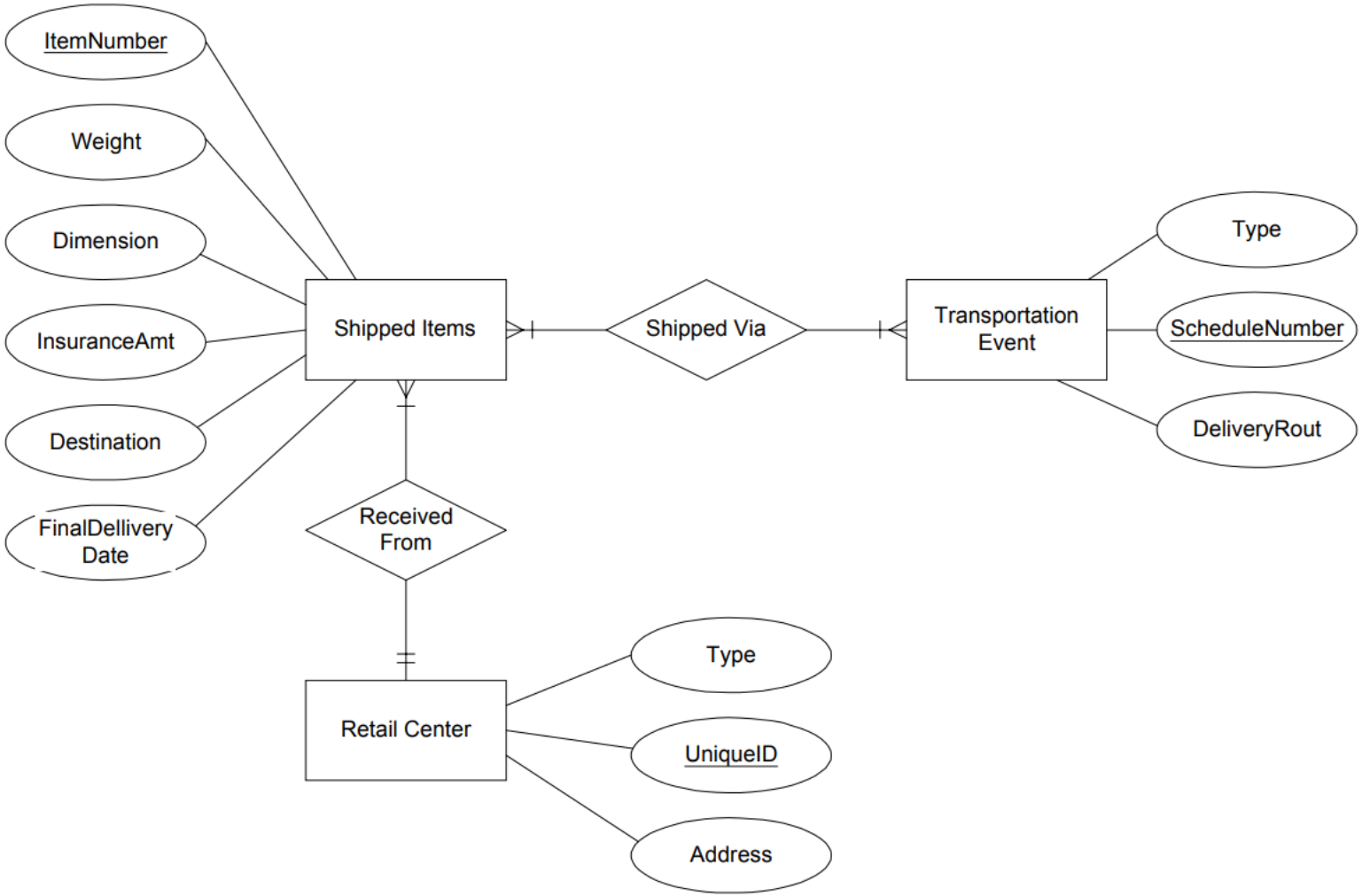


Figure 2: An ER diagram for a BANK database schema (ref: Fundamental of Database Systems by Elmasri and Navathe)

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute.

Please create an Entity Relationship diagram that captures this information about the UPS system. Be certain to indicate identifiers and cardinality constraints.


ItemNumber
Weight
Dimension
InsuranceAmt
Destination
FinalDellivery Date
Shipped Items
Shipped Via
Transportation Event
Type
ScheduleNumber
DeliveryRout
Received From
Retail Center
Type
UniqueID
Address

Extended ER Diagrams

- Enhanced Entity-Relationship Modelling
  - Superclass / subclass
  - Specialization and Generalization

Enhanced ER (EER) Modelling

- Since the 1980s there has been an increase in the emergence of new database applications with more demanding requirements.

- Basic concepts of ER modelling are not sufficient to represent the requirements of the newer, more complex applications.

- Response is development of additional ‘semantic’ modelling concepts.

- Semantic concepts are incorporated into the original ER model and is called the Enhanced Entity-Relationship (EER) model.
  - That is EER includes all modelling concepts in ER

EER Modelling

- EER is also Known as Extended ER model

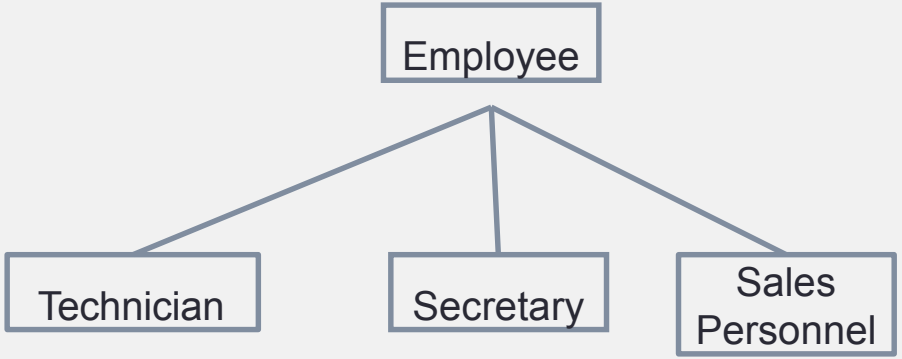
- Created to design more accurate database schemas
- Reflect the data properties and constraints more precisely

- Additional concepts of EER model includes specialization / generalization, and categorization.

Superclass and Subclass

- A **Superclass** – is an entity type that includes distinct Subclasses that require to be represented in a data model.

- A **Subclass** is an entity type that has a distinct role and is also a member of the Superclass



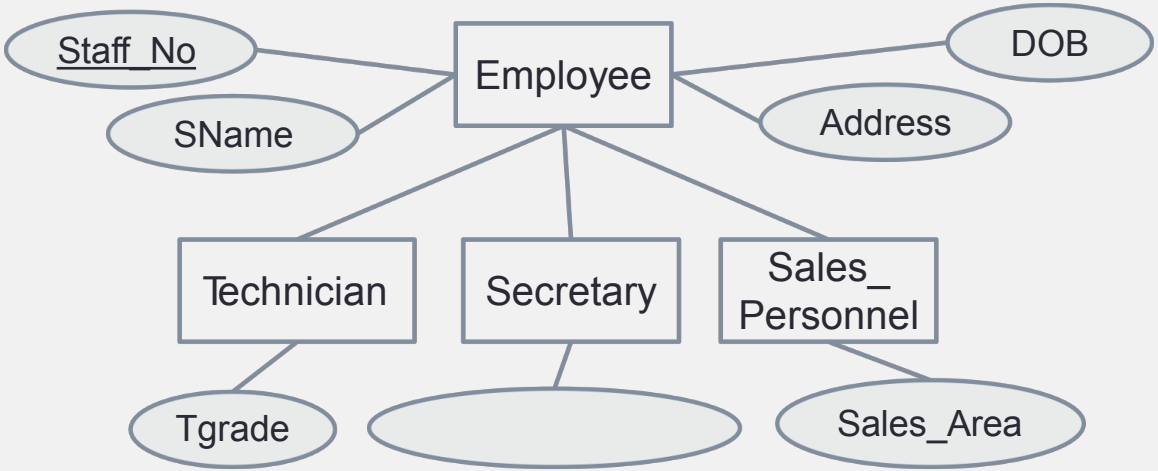
Superclass and Subclass Examples

- Member (superclass): Life_Member, Regular_Member, Season_Member (subclasses)
- Staff (superclass): Full_Time_Staff, Part_Time_Staff (subclasses)

- It adds more semantic clarity
  - If only Full_Time_Staff belongs to the Welfare_Society then it can be represented as
    - *Belongs_To :* <Full_Time_Staff, Welfare_Society>
    - **Not as** *Belongs_To :* <Staff, Welfare_Society>

Inheritance

- Subclass entity inherits <u>all attributes and relationships</u> of superclass
- **Attribute Inheritance**
  - An entity in a Subclass may possess subclass specific attributes, as well as those associated with the Superclass.



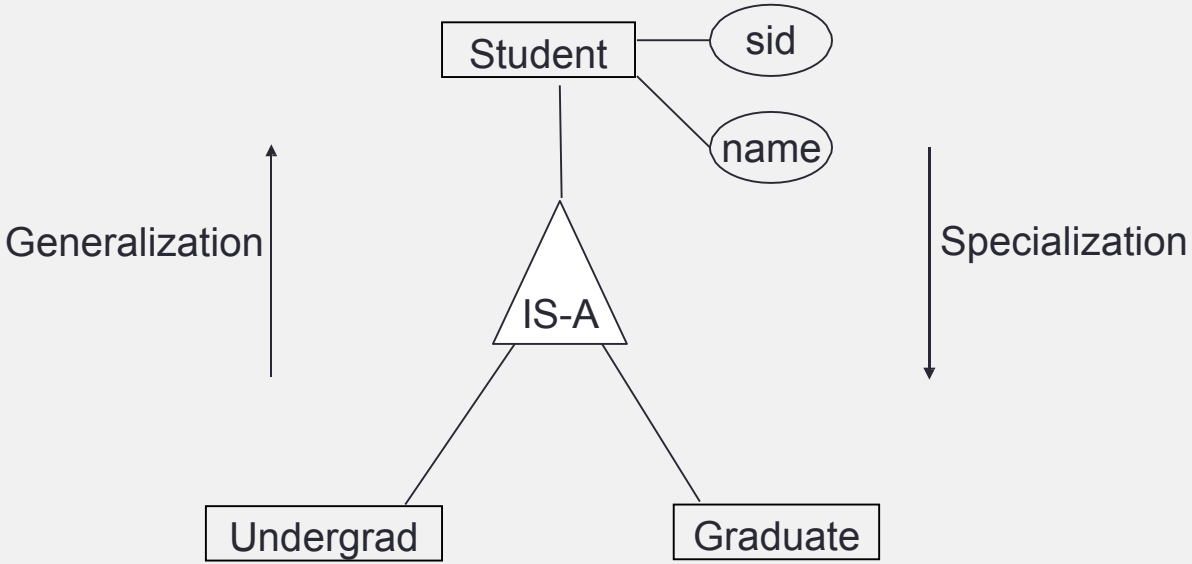
IS-A (‘is a’)Hierarchies

- Inheritance IS-A (instance) Relationship
- If we declare A **is a** B. Every A entity is considered to be a B entity.
- Secretary **IS-A** Employee

- Single inheritance results in a hierarchy (tree structure)
  - Every subclass has only one superclass

- Multiple inheritances results in a lattice
  - Subclass with more than one superclass
  - E.g. Student ❼Student_Assistant <- Employee

Is-A Relationship



Specialization and Generalization

- **Specialization**
  - The process of maximizing the differences between members of an entity by identifying their distinguishing characteristics.
  - Identify subclasses and their *distinguishing characteristics* (attributes, relationships)
  - Start with entity type then define subclasses by successive specialization

Top-Down

- **Generalization**
  - The process of minimizing the differences between entities by identifying their common characteristics.
  - Aggregates entities into a superclass entity types *by identifying their common characteristics*

Bottom up

Specialization and Generalization

- CAR, TRUCK generalized into VEHICLE;

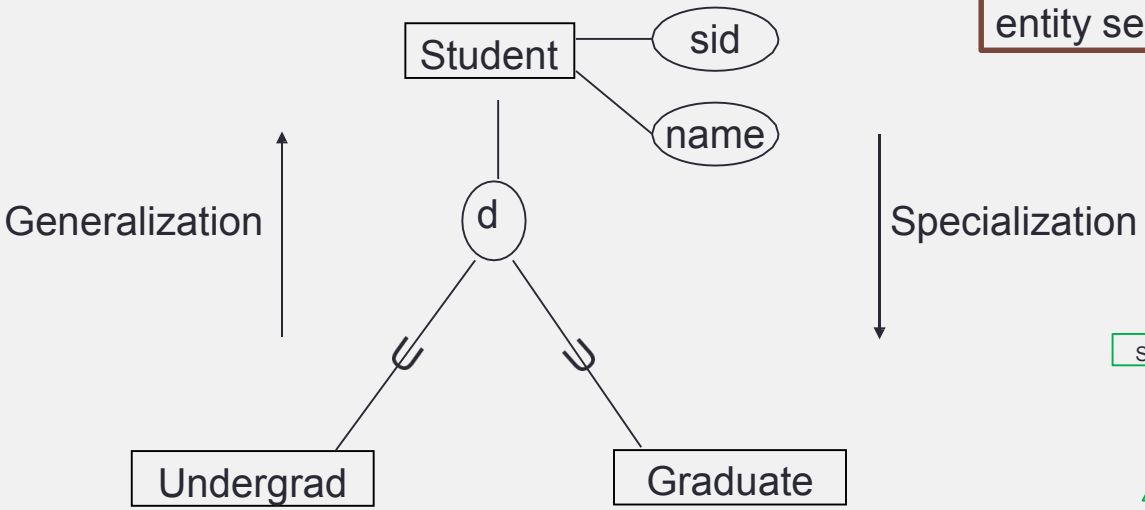
- Both CAR, TRUCK become subclasses of the superclass VEHICLE.
  - We can view {CAR, TRUCK} as a specialization of VEHICLE

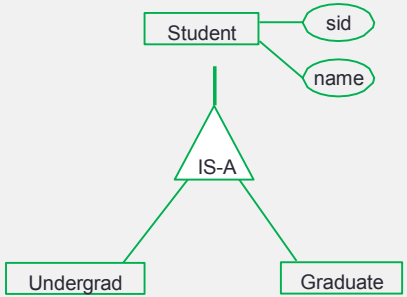
  - Alternatively, we can view VEHICLE as a generalization of CAR and TRUCK

Specialization and Generalization

An alternative notation is the Union symbol

The circle with d specifies that the specializations are disjoint. A member of Undergrad entity set NOT a member of the graduate entity set.





Reasons for Specialization?

- Certain attributes may apply to some but not all entities of the superclass
  - E.g. Secretary – typing speed
  - Thus, it minimizes Null Values

Specific (local) attributes of subclass

- Some relationship types may be participated in only by entities that are members of the subclass
  - If only Full_Time Employee belongs to Welfare_Society

Specific relationship

Specialization and Generalization

Specialization and generalization has:

- Disjointness (Inclusion) Constraints
- Completeness (Participation) Constraints

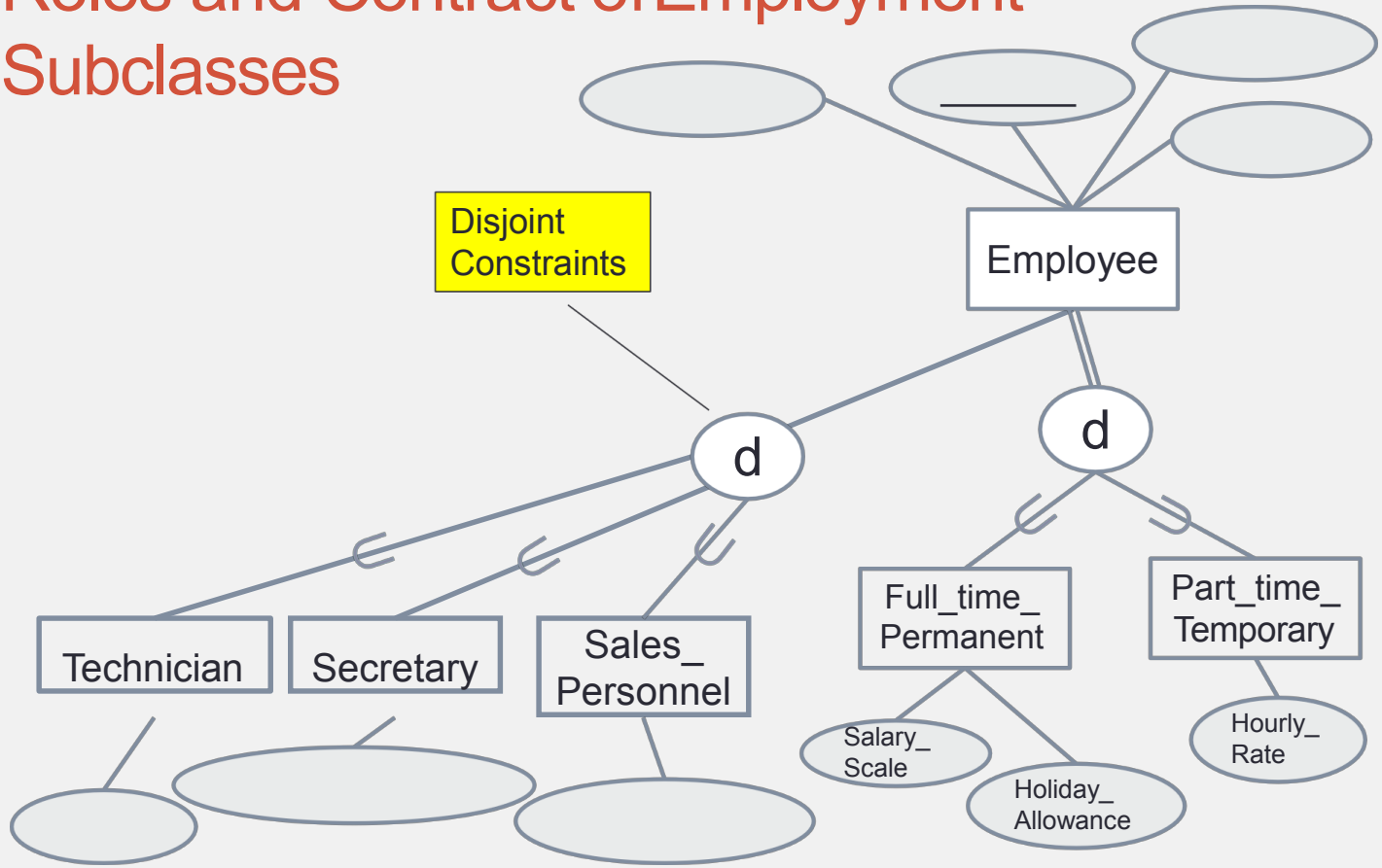
- Disjointness and Completeness constraints are independent

Disjointness Constraint

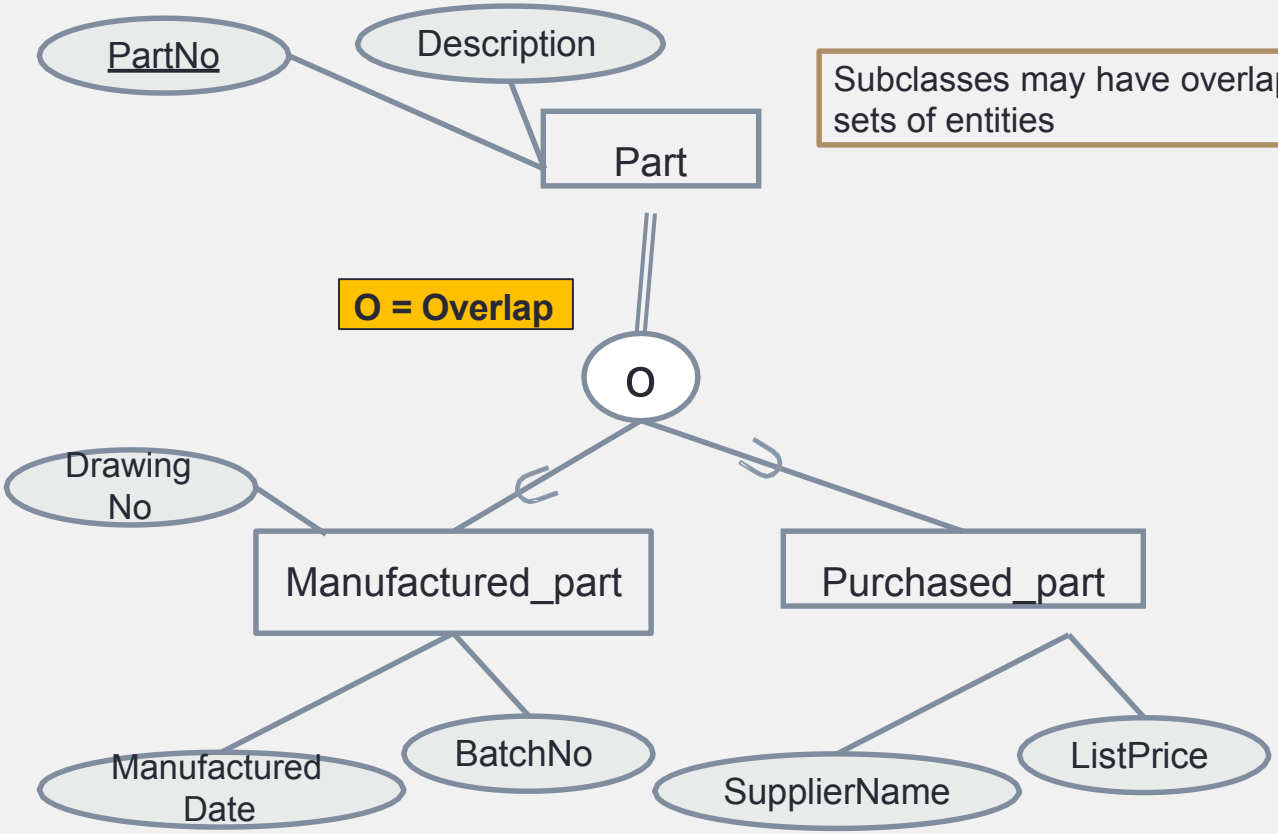
- Disjoint
  - Members in different Subclasses from the same Superclass are completely different.
  - An entity can be a member of at most one of the subclasses of the specialization
  - Specified by **d** in EER diagram

- Non-Disjoint (overlap)
  - Members in a Superclass can be enrolled in more than one Subclass
  - That is the same entity may be a member of more than one subclass of the specialization
  - Specified by **o** in EER diagram

Specialization of Employee Entity into Job Roles and Contract of Employment Subclasses



Non-Disjoint Constraints



Subclasses may have overlapping sets of entities

Completeness Constraints

- **Total specialization**
  - All member in the Superclass must participate in either one Subclass.
  - Shown in EER diagrams by a double line

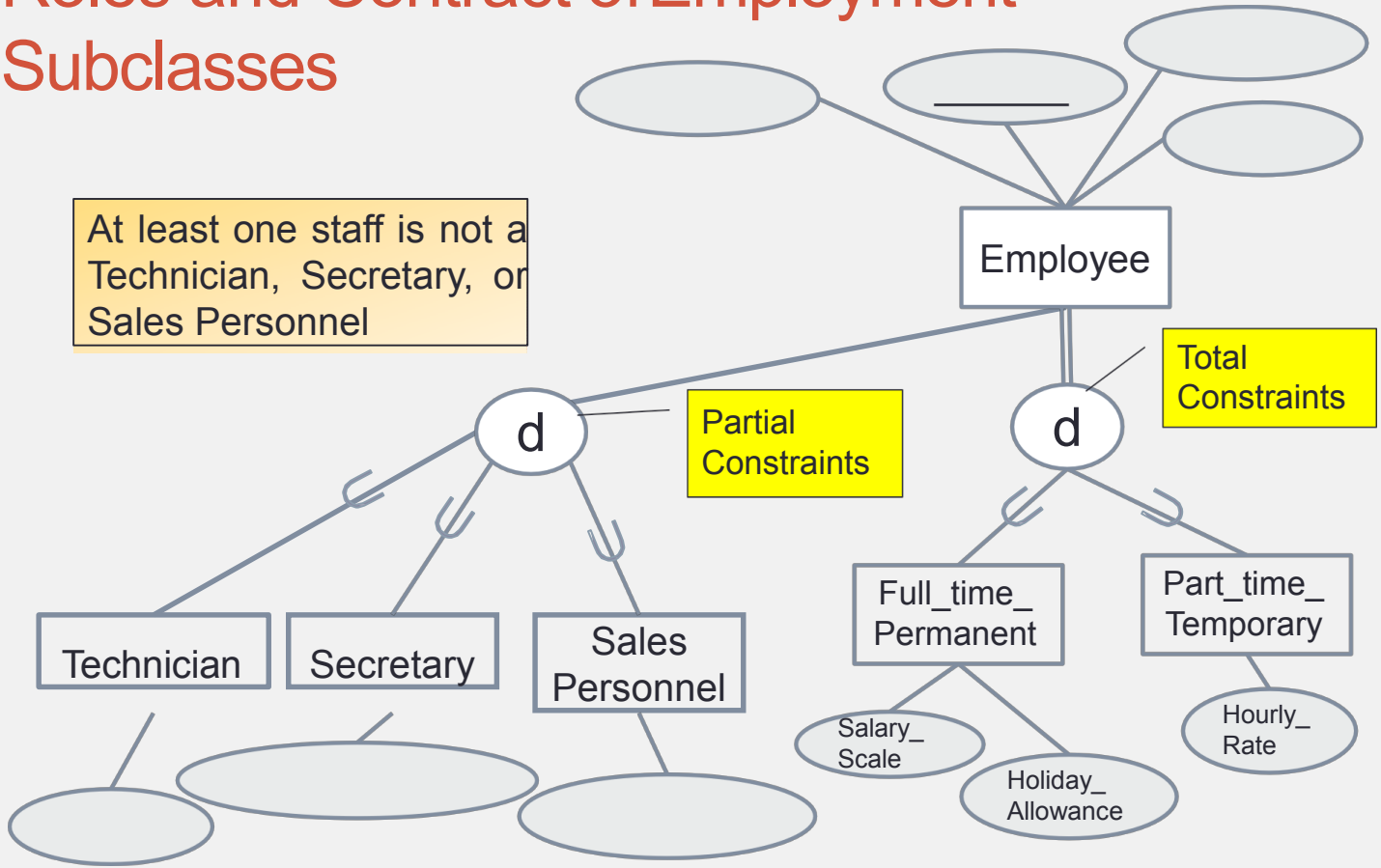
- **Partial specialization**
  - At least one member in the Superclass does not participate in the

Subclass.

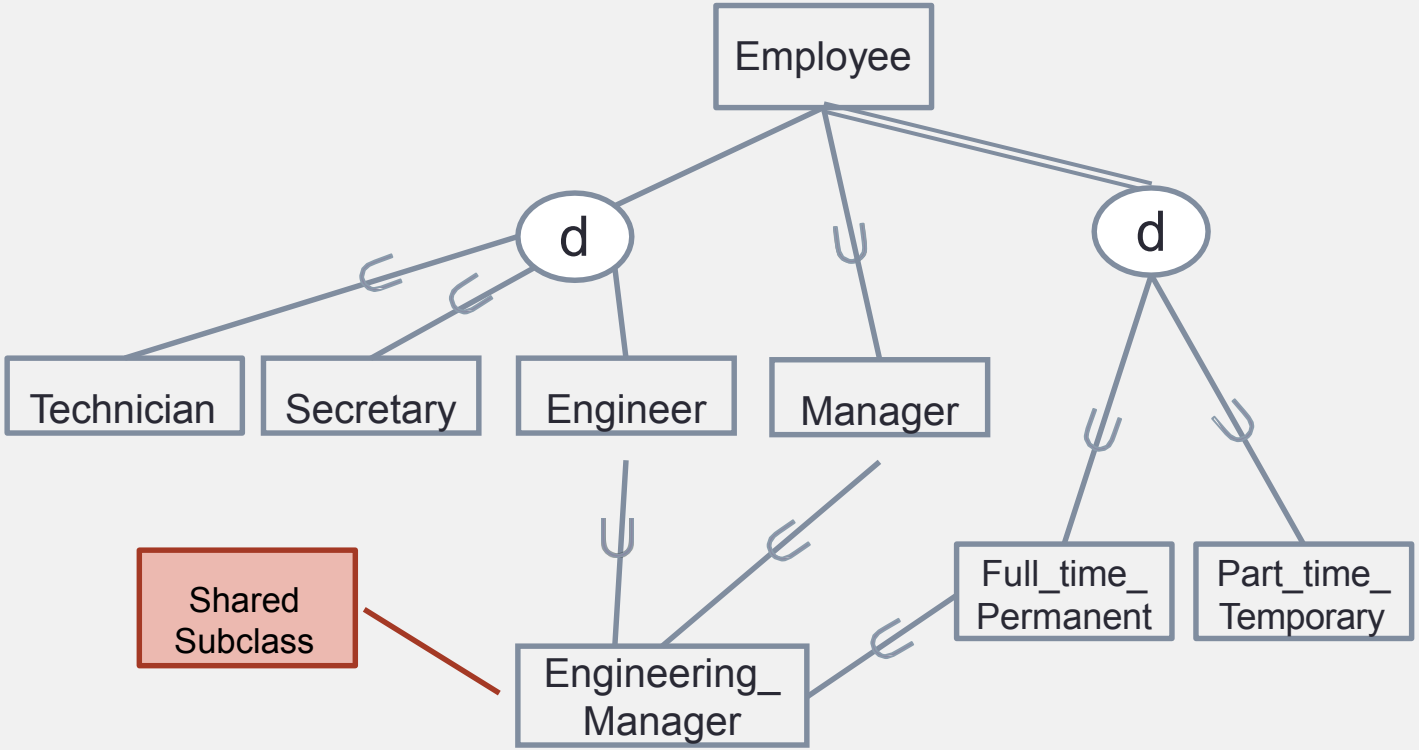
  - Partial allows an entity not to belong to any of the subclasses
  - Shown in EER diagrams by a single line

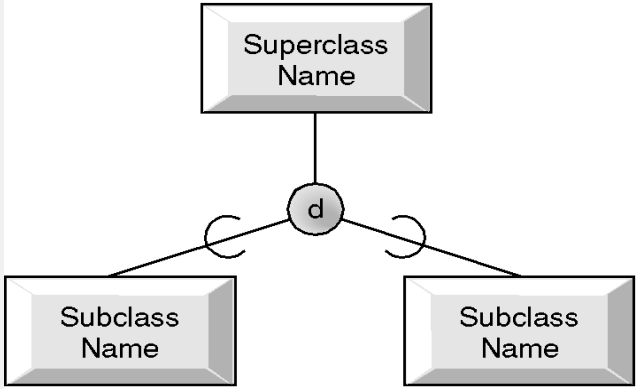
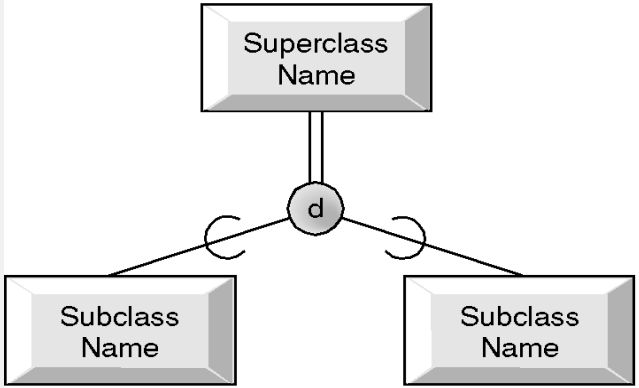
- Generalization usually is total because the superclass is derived from the subclasses.

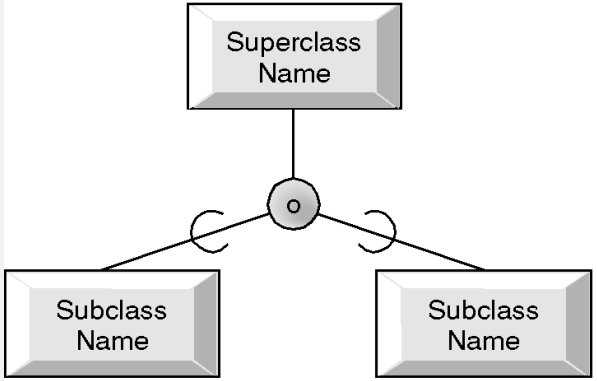
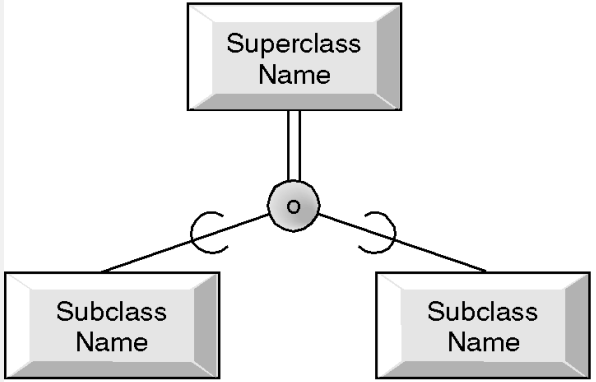
Specialization of Employee Entity into Job Roles and Contract of Employment Subclasses



Specialization Lattice (Multiple Inheritance)



Notation	Meaning
	Superclass/Subclass Relationship Disjoint, Partial
	Superclass/Subclass Relationship Disjoint, Total

Notation	Meaning
	Superclass/Subclass Relationship Overlapping, Partial
	Superclass/Subclass Relationship Overlapping, Total

Building EER

- Identify entity types.
- Identify relationship types.
- Determine cardinality and participation constraints of relationship types.
- Identify and associate attributes with entity or relationship types.
- Determine candidate and primary key attributes.
- Specialize / generalize entity types.
- Draw the EER Diagram.